

2(a)(i)	one mark each correct row <table><tr><th>cell structure</th><th>function</th></tr><tr><td>nucleolus ;</td><td>manufactures ribosomal subunits from proteins and ribosomal RNA</td></tr><tr><td>smooth endoplasmic reticulum ; R endoplasmic</td><td>synthesises triglycerides and other lipids</td></tr><tr><td>pair of centrioles ; R centrosomes</td><td>organise microtubules of the cell cytoskeleton</td></tr></table>	cell structure	function	nucleolus ;	manufactures ribosomal subunits from proteins and ribosomal RNA	smooth endoplasmic reticulum ; R endoplasmic	synthesises triglycerides and other lipids	pair of centrioles ; R centrosomes	organise microtubules of the cell cytoskeleton	3
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2(a)(ii)	structures should be labelled and drawn with enough detail not to be mistaken for another structure all three correct = 2 marks one or two correct = 1 mark all three structures correctly drawn but no labels = 1 mark - roughly spherical organelle within the nucleus labelled as nucleolus - minimum of one single membrane-bound tubular sac labelled as, smooth endoplasmic reticulum / smooth ER / SER R if dots (ribosomes) drawn and resembles RER - two, short lines or cylinders, not parallel to each other labelled as (pair of) centrioles R if inside the nucleus	2								

Question	Answer	Marks
2(b)(i)	<i>allow, M. tuberculosis / pathogens / bacteria, for bacterial cells</i> endocytosis occurs / described ; e.g. (macrophage cell surface) membrane, surrounds / AW, bacterial cells pseudopodia, surrounds / form round / AW, bacterial cell (alveolar) macrophage / phagocyte / (phagocytic) cell, envelops / engulfs, bacterial cells ; event occurring before <u>or</u> after ; <i>e.g. before</i> <i>ref. to</i> chemotaxis / chemotactic response bacterial cells / bacteria / (bacterial) antigens, bind / attach / join / AW, to (macrophage cell surface), receptors / membrane I binding to, macrophage / cell <i>suggestion of</i> (in alveolar space) opsonisation of /antibody binding to, bacterial cells <i>e.g. after</i> membrane fusion / (phagocytic), vacuole pinches off / AW A vesicle for vacuole	2
2(b)(ii)	lysosomes / lysosome ; A lysosomal vesicles	1

Question	Answer	Marks
2(c)	any four from: 1 cells are replicating / AW (because live vaccine) ; 2 high / increased / AW, levels of (non-self / foreign) antigen (because live vaccine) ; 3 (so) strong / effective / good / AW, (primary) <u>immune response</u> ; A described e.g. (T-/B-) memory cells are formed <i>idea that</i> memory cells present when <i>Mycobacterium</i> infects / AW secondary response occurs when <i>Mycobacterium</i> infects / AW 4 provides (artificial) active immunity ; R natural active immunity 5 (so) boosters not needed (to build sufficient immunity) ; 6 vaccine does not cause, disease / TB (because, weakened / attenuated) ; I doesn't show symptoms 7 easy to, transport / store / deliver (because freeze-dried) ; 8 AVP ;; e.g. effective against <i>M. tuberculosis</i> and <i>M. bovis</i> no / few / mild, side effects, so people not discouraged (from having vaccine) memory cells / immunity, (relatively) long-lived / long-lasting only need one dose (good immunity for most individuals) provides herd immunity (<i>a consequential effect</i>)	4